Answer Key - Mathematics - Undergraduate Level Assessment 10 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** B
   Options: A) (0, 0, 1); B) (3/7, 3/14, 9/14); C) (7/15, 8/15, 1/15); D) (5/6, 1/3, 1/3)
   Note: Correct option: B - (3/7, 3/14, 9/14)

2. **Answer:** C
   Options: A) True, True; B) False, False; C) True, False; D) False, True
   Note: Correct option: C - True, False

3. **Answer:** D
   Options: A) 1; B) 3; C) 5; D) 7
   Note: Correct option: D - 7

4. **Answer:** D
   Options: A) 1; B) i; C) 32; D) 32 i
   Note: Correct option: D - 32 i

5. **Answer:** C
   Options: A) G is abelian; B) G = {e}; C) a^3 = e; D) a^2 = a
   Note: Correct option: C - a^3 = e

True / False

6. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: 'f(log c)'.

7. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

8. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

9. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: 'infinite, non abelian group'.

10. **Answer:** True
    Options: A) True; B) False
    Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** Suppose our polynomial is equal to

$$ax^3 + bx^2 + cx + d$$

Then we are given that

$$-9 = b + d - a - c.$$

If we let $-a = a' - 9, -c = c' - 9$ then we have

$$9 = a' + c' + b + d.$$

This way all four variables are within 0 and 9. The number of solutions to this equation is simply 123 = 220 by stars and bars.
Note: Students should show all steps in their mathematical solution.

12. **Answer:** Converting to degrees,

$$-\frac{3\pi}{4} = \frac{180^\circ}{\pi} \cdot \left(-\frac{3\pi}{4}\right) = -135^\circ.$$

Since the tangent function has period 180°, $\tan(-135^\circ) = \tan(-135^\circ + 180^\circ) = \tan 45^\circ = 1$.
Note: Students should show all steps in their mathematical solution.

End of Answer Key